Introductory Econometrics with Recitation: TA Session Week 6

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Today's agenda

1. Data Wrangling
   - `nrow()`, `ncol()`, `dim()`
   - `nchar()`
   - `substring()`
   - `grepl()`
2. Summary Statistics Table
   - `table()`
   - `addmargins()`
3. Multiple Regression
   - `lm_robust()`
   - `screenreg()`

Today's Dataset

- Today we will use the following CSV file from the OpenIntro Website:
  - `ncbirths.csv`: A 2004 dataset from North Carolina on births, including mothers' habits and practices, with 1,000 cases.
  - `loan_full_schema.csv`: The dataset represents thousands of loans made through the Lending Club platform.
- Please import the following datasets into RStudio.

```
loan <- read.csv("loans_full_schema.csv")
birth <- read.csv("data/ncbirths.csv")
```

Number of rows/columns: nrow(), ncol(), dim()

- **Syntax:**

```
nrow(df) # or matrix
ncol(df)
dim(df)
```

- These functions work on dataframes or matrices, but not on vectors.
- **Example:**

```
df <- data.frame(
  name  = c("Alice", "Bob"),
  score = c(80, 90),
  pass  = c(TRUE, TRUE)
  )

nrow(df)  # 2
ncol(df)  # 3
dim(df)   # 2 3
```

Count Number of Characters: nchar()

- nchar() counts the number of characters in a string.
- **Syntax:**

```
nchar(x)
```

- **Example:**

```
word1 <- "apple"
nchar(word1)
# [1] 5

words <- c(15, -3.5, TRUE)
nchar(words)
# [1] 2 4 1
```

Trim String: substring()

- **Syntax:**

```
substring(str, start_index, end_index)
```

- **Example:**

```
str1 <- "Cat_and_Dog"
substr1 <- substring(str1, 1, 3)
print(substr1)
```

- **Output:**

```
[1] "Cat"
```

Find patterns inside element: grepl()

- **Syntax:**

```
grepl(pattern, str)
grepl(pattern, str_vec)
```

- grepl() will return a logical vector: TRUE if the pattern is inside that element, FALSE otherwise.
- **Example:**

```
grepl("app", "apple") # TRUE

fruits <- c("apple", "banana", "pineapple", "pear")
grepl("apple", fruits) # TRUE FALSE TRUE FALSE

strings <- c("abcba", "acbcb", "abbbc", "babcc")
grepl("abc", strings) # TRUE FALSE FALSE TRUE
```

Create Contingency Table: `table()`

	home ownership		
application type	**Rent**	**Mortgage**	**Own**
individual	3496	3839	1170
joint	362	950	183

Table: Homeownership by application type

Create Contingency Table: table()

- **Syntax:**

```
# dnn = dimension names
A <- table(..., dnn = ...)
```

- **Example:**

```
loan <- read.csv("loans_full_schema.csv")
table <- table(loan$application_type,
               loan$homeownership,
               dnn = c("application type",
                       "home ownership"
                       )
               )
```

Add Sum Row and Sum Column: `addmargins()`

application_type	homeownership			
	Rent	Mortgage	Own	Total
individual	3496	3839	1170	8505
joint	362	950	183	1495
Total	3858	4789	1353	10 000

Table: Homeownership by application type, with sum row and sum column

Add Sum Row and Sum Column: addmargins()

- **Syntax:**

```
A <- addmargins(table)
```

- **Example:**

```
table_withsum <- addmargins(table)
```

Heteroscedasticity-Robust Standard Errors: `lm_robust()`

- **Homoscedasticity** (A5) is often violated, that is,

$$\mathrm{Var}(u_i \mid X_i) \text{ depends on } X_i$$

- $\hat{\beta}_1$ is still unbiased and consistent; however, when estimating $\mathrm{Var}(\hat{\beta}_1)$, we need a heteroscedasticity-robust estimator:

$$\hat{\sigma}^2_{\hat{\beta}_1} = \frac{\frac{1}{n} \times \frac{1}{n-2} \sum_{i=1}^{n} (X_i - \bar{X})^2 \hat{u}_i^2}{\left[\frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2\right]^2}$$

- This can be implemented by the `lm_robust()` function.
- No need to memorize the formula, just interpret it.

Heteroscedasticity-Robust Standard Errors: lm_robust()

- **Syntax:**

```
library(estimatr) # import the package!

lm_model_robust <- lm_robust(
  data = myData,
  Y ~ X1 + X2 + X3
)
```

- **Example:**

```
lm_model_multiple_robust <- lm_robust(
  data = birth,
  weight ~ weeks + gender + visits
)
```

Heteroscedasticity-Robust Standard Errors: lm_robust()

- **Output:**

```
> summary(lm_model_multiple_robust)

Call:
lm_robust(formula = weight ~ weeks + gender + visits, data = birth)

Standard error type:  HC2

Coefficients:
             Estimate Std. Error t value   Pr(>|t|) CI Lower CI Upper  DF
(Intercept) -6.208828   0.532882 -11.651  1.745e-29 -7.25454 -5.16312 986
weeks        0.339617   0.013970  24.311 1.163e-102  0.31220  0.36703 986
gendermale   0.365473   0.070477   5.186  2.611e-07  0.22717  0.50377 986
visits       0.009322   0.009229   1.010  3.127e-01 -0.00879  0.02743 986

Multiple R-squared:  0.4492 ,  Adjusted R-squared:  0.4476
F-statistic: 216.8 on 3 and 986 DF,  p-value: < 2.2e-16
```

Heteroscedasticity-Robust Standard Errors: lm_robust()

- **Output using lm():**

```
> summary(lm_model_multiple)

Call:
lm(formula = weight ~ weeks + gender + visits, data = birth)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.7393 -0.6886 -0.0053  0.6802  4.2551 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -6.208828   0.474458 -13.086  < 2e-16 ***
weeks        0.339617   0.012485  27.201  < 2e-16 ***
gendermale   0.365473   0.070411   5.191 2.55e-07 ***
visits       0.009322   0.009045   1.031    0.303    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.107 on 986 degrees of freedom
Multiple R-squared:  0.4492,^^IAdjusted R-squared:  0.4476 
F-statistic: 268.1 on 3 and 986 DF,  p-value: < 2.2e-16
```

Regression Table: screenreg()

- **Syntax:**

```r
library(texreg) # import the package!

screenreg(
  list(model1, model2, ...),
  custom.model.names =
    c("Model 1", "Model 2", ...),
  include.ci = FALSE,
  digit = 3
)
```

Regression Table: screenreg()

- **Example:**

```
screenreg(
  # models you want to present in table
  list(lm_model, lm_model_multiple, lm_model_multiple_robust),

  # names for each model
  custom.model.names =
    c("Model 1: Single",
      "Model 2: Multiple",
      "Model 3: Multiple (Robust SE)"),

  # present standard error
  include.ci = FALSE,
  digit = 3
)
```

Regression Table: screenreg()

- **Result**

```
=============================================================================
               Model 1: Single  Model 2: Multiple  Model 3: Multiple (Robust SE)
-----------------------------------------------------------------------------
(Intercept)     -6.095 ***       -6.209 ***          -6.209 ***
                (0.465)          (0.474)             (0.533)
weeks            0.344 ***        0.340 ***           0.340 ***
                (0.012)          (0.012)             (0.014)
gendermale                        0.365 ***           0.365 ***
                                 (0.070)             (0.070)
visits                            0.009               0.009
                                 (0.009)             (0.009)
-----------------------------------------------------------------------------
R^2              0.449            0.449               0.449
Adj. R^2         0.448            0.448               0.448
Num. obs.          998              990                 990
RMSE                                                  1.107
=============================================================================
*** p < 0.001; ** p < 0.01; * p < 0.05
```